KSHITIZ RAY

MECHATRONICS ENGINEERING STUDENT | MECHANICAL DESIGN · ROBOTICS

Jamshedpur, Jharkhand · Studying in Bhubaneswar, Odisha

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PROFILE

B.Tech Mechatronics student at KIIT with a hands-on diploma from NTTF and industry experience at Tata Steel and COTMAV Robotics & Automation. Interested in mechanical design, robotics integration, product development, and smart manufacturing. Combines CAD modeling, mechanical assembly, and embedded development through practical engineering projects.

INDUSTRY EXPERIENCE

Tata Steel — Engineering Internship | May–June 2026

- Worked on an automation concept for unloading desulphurization (DS) material from tanker trucks, with an emphasis on reducing manual hose handling and improving operator safety.
- Explored mechanical design, robotic handling, and industrial sensing for the unloading workflow.

COTMAV Robotics & Automation — Junior Project Engineer Intern | Sept 2024–Mar 2025

- Assisted FANUC industrial robot implementation; supported installation, mechanical setup, testing, and commissioning in manufacturing environments.
- Contributed to robotic welding and spray-painting automation applications, with exposure to client projects including Endurance, Studds Helmets, Highco Engineering Works, Arvind Mill, and RKFL.

EDUCATION

B.Tech, Mechatronics Engineering — KIIT University, Bhubaneswar | 2025–Present

Diploma, Mechatronics Engineering — NTTF R.D. Tata Technical Education Centre, Jamshedpur | 2021–2024 | CGPA: 7.82

SELECTED ENGINEERING PROJECTS

- **4-DOF SCARA Robotic Arm:** Designed the mechanical layout, joints, and actuator arrangement for a four-degree-of-freedom robotic arm.
- **Delta 3D Printer:** Designed and assembled a delta-configuration printer; tuned slicing and print settings in Ultimaker Cura.
- **ESP32 Smart Home Automation:** Developed Wi-Fi-enabled appliance control through a mobile interface.
- **IoT Weather Monitoring:** Built a microcontroller-based system to collect and report sensor-based environmental data.

TECHNICAL TOOLKIT

Mechanical & manufacturing: SolidWorks, Siemens NX, Fusion 360, MasterCAM, CAD modeling, mechanical assembly, MIG welding, additive manufacturing, Ultimaker Cura.

Robotics & automation: FANUC ARC Mate, FANUC LR Mate, Elite Cobots, robot integration, industrial automation, robotic welding and spray-painting applications.

Embedded & software: Arduino, ESP32, Raspberry Pi, Embedded C, IoT, temperature/ultrasonic/IR sensors, reed switches, Arduino IDE, Tinkercad. PLC: currently learning.

QUALIFICATIONS & RECOGNITION

NOCN Level 5 Diploma in Mechanical Engineering — Mechatronics & Smart Manufacturing (**Merit**); NSDC Technician Instrumentation (Level 4); NSDC Wireman Control Panel Electronics (Level 3).

Autodesk Skill Surge Inter-State Design Challenge 2025 — **Finalist**; NTTF Home Automation achievement; Student AI Masterclass participation (2026); Tata Steel Digi-Shala courses (2026).